Week 10 – Workshop
Geothermics: Finite Differences

Adelaide University

Geophysics 3A: Potential Fields & Geothermics

July 31, 2026

Preparation

Pre-reading: Course Notes, Ch. 18
Lecture videos:
Notes:

Purpose

This workshop builds intuition for finite difference methods through physical reasoning and interactive activities. Students first derive the heat equation from energy conservation, then experience how numerical models work by acting as nodes that update temperature based on their neighbors. The workshop then explores how boundary conditions represent geological assumptions and how numerical stability reflects limits on how heat can physically propagate. By the end, students should understand that finite difference methods are simply local energy balance, that heat flow emerges from neighbor interactions, and that both boundary conditions and stability have direct physical meaning.

Learning Outcomes

Students will:

- Understand finite difference equations as local statements of energy conservation.
- Develop intuition for how numerical solutions evolve through neighbor interactions.
- Interpret boundary conditions as geological assumptions.
- Understand numerical stability as a limit on physical information transfer.

Session Flow (?? min)

0–10 min: Warm-up — What breaks down?

Think back over the analytic solutions in Weeks 8 and 9 (steady-state geotherm, half-space cooling, plate cooling). Discuss a geological situation where each analytic solution would fail. What assumption(s) breaks down?

10–25 min: Mini-lecture — From energy conservation to finite differences

- The heat equation as a statement of local energy balance on a small rock volume.
- Discretization: replacing continuous derivatives with differences between neighboring nodes.
- The finite difference stencil: each node updates based on its neighbors.
- Stability: why the time step cannot be arbitrarily large ($\Delta t < \Delta x^2 / 2\kappa$).

What makes geological systems difficult to solve analytically? Consider:

- Analytic solutions require simplifications:
  - constant properties
  - simple geometries
  - simple boundary conditions
- Real geology violates these assumptions.

25–55 min: Activity 10.A: From Conservation of Energy to Finite Differences

Finite difference equations arise from applying conservation of energy to small rock volumes.

- Nodes represent local rock volumes.
- Heat flux enters and leaves each node.
- Heat production adds energy locally.

The equation is *not a numerical trick*. The Laplacian as a measure of curvature (difference from neighbors). This is just bookkeeping of energy.

55–85 min: Activity 10.B: Human Finite Difference Solver

Numerical models evolve through local interactions between neighboring nodes. We'll do an exercise with you as the computer to see how finite difference solutions work.

We'll explore

- smoothing of temperature
- spreading of heat
- role of gradients

85–95 min: Break

95–125 min: Activity 10.C: Boundary Conditions as Geological Assumptions

Boundary conditions determine what physical system the model represents. There are two fundamental types:

- Fixed value (Dirichlet)
- Fixed flux (Neumann)

We'll also look at two additional types (Robin and Cauchy) that are combinations or independent applications of these types.

125–150 min: Activity 10.D: Stability and Information Transfer

Stability limits impose a "speed limit" on thermal information:

$$\Delta t < \frac{\Delta x^2}{2\kappa}$$

Heat propagates gradually through diffusion—numerical schemes must respect this physical constraint. We'll explore:

- small vs large time steps
- what happens when updates are too large
- stability = causality
- information cannot skip nodes

Instability is not just numerical—it is physically impossible behavior.

150–160 min: Wrap-up

Key connections

- Finite differences = local energy balance + neighbor interaction
- Boundary conditions = geological assumptions
- Stability = physical constraint on information transfer

Discussion

- How does the numerical method compare to analytic solutions?
- What advantages do numerical models provide?
- Where might these methods be applied beyond heat flow?

160–180 min: Course Conclusion

This is the final workshop of the course. Rather than a preview, use this time to connect ideas across all ten weeks.

Fields and gradients (Week 1): all geophysical observations are measurements of fields; gradients tell us where physical property contrasts occur.

Physical properties (Week 2): the link between measurement and geology lives in physical properties and their mixing laws.

Inversion (Week 3): going from data to model requires confronting non-linearity, non-uniqueness, and the role of prior information.

Gravity and isostasy (Weeks 4–5): density contrasts at depth shape both the gravity field and the long-term mechanical balance of the lithosphere.

Magnetics and Fourier methods (Weeks 6–7): vector fields, polarity, depth–wavelength relationships, and spectral filtering.

Geothermics (Weeks 8–9): heat flow connects surface measurements to deep thermal structure via conduction, diffusion, and phase changes.

Finite differences (Week 10): numerical methods let us move beyond analytic limits to model the real, heterogeneous Earth.

Workshop Notes

Workshop Notes

Activity 10.A
From Conservation of Energy to Finite Differences

Purpose

To understand finite difference heat flow equations as direct consequences of energy conservation applied to small volumes of rock.

Learning Outcomes

By the end of this activity, you should be able to:

- Derive the finite difference heat equation from energy conservation applied to a discrete node.
- Interpret the Laplacian finite difference stencil $(T_{i+1} - 2T_i + T_{i-1})$ as a measure of local curvature or departure from the mean of neighbors.
- Explain the role of heat production A_i in modifying the temperature update.
- Connect the discrete finite difference equation to the continuous partial differential equation it approximates.

Consider a vertical column of sediment divided into small layers. Heat flows between layers, and each layer can gain or lose energy depending on its neighbors and any internal heat production.

Rather than solving a continuous equation, we will track how heat moves between *discrete points*.

Part A: Physical intuition

- If a layer is hotter than the ones above and below it, does it gain or lose heat?
- If heat is being produced inside the layer, what happens over time?

Part B: Setting up the system

- Sketch a one-dimensional grid of nodes labeled T_{i-1}, T_i, and T_{i+1}.
- Indicate the spacing Δz.

Part C: Heat flux

Heat flows according to Fourier's law:

$$q_z = -k\frac{\partial T}{\partial z} \tag{1}$$

Tasks:

- Write an expression for heat flowing *into* node i from Equation **??**.
- Write an expression for heat flowing *out of* node i from Equation **??**.

Part D: Energy conservation

Energy conservation for a small volume gives:

$$\text{Rate of change} = \text{in} - \text{out} + \text{produced}$$

Task: Combine your expressions to derive:

$$\frac{\partial T_i}{\partial t} = \kappa\frac{T_{i+1} - 2T_i + T_{i-1}}{\Delta z^2} + \frac{A_i}{\rho c_p} \tag{2}$$

Part E: Interpretation

Explain in words:

- What does the term $T_{i+1} - 2T_i + T_{i-1}$ represent?
- What role does heat production A_i play?
- What happens if $A_i = 0$?

Key Ideas

Finite difference equations are not numerical tricks—they are simply energy conservation applied locally.

Finite differences are discretized energy conservation. The finite difference heat equation is not an approximation introduced for computational convenience — it is the exact energy balance for a discrete rock volume, with the continuous derivatives replaced by differences between neighboring nodes. This is the same physics as the heat equation; only the representation changes.

The Laplacian stencil measures curvature. The term $T_{i+1} - 2T_i + T_{i-1}$ is the discrete second derivative of temperature. It is positive when a node is cooler than the average of its neighbors (it receives net heat) and negative when it is warmer (it loses heat). This is exactly how diffusion works: it smooths by removing curvature.

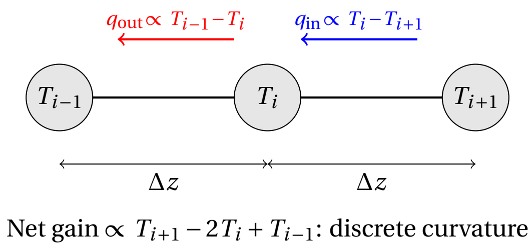


Figure 1: Heat enters node i from the left (red) and exits to the right (blue). The net heat gain is proportional to the curvature of the temperature profile.

The method scales to any geometry. Unlike analytic solutions, which require special symmetries, the same finite difference update rule applies to any grid, any set of boundary conditions, and any spatially variable material properties. This generality is the key advantage of numerical methods.

Activity 10.B
Human Finite Difference Solver

Purpose

To develop intuition for finite difference methods by acting as the nodes in a numerical heat flow model.

In a numerical model, each point (node) updates its temperature using only information from its neighbors.

In this activity, you will act as the nodes in a finite difference model.

Learning Outcomes

By the end of this activity, you should be able to:

- Apply the finite difference update rule by computing temperature from neighboring values.
- Predict qualitatively how an initial temperature distribution evolves under the diffusion update rule.
- Explain why temperature peaks smooth out and why the system approaches a uniform (or linear) steady state.
- Describe why heat transfer is a local process and how information about temperature propagates through a grid one node at a time.

Box .1: Human Computer

Before electronic computers existed, a *computer* was literally a person employed to carry out calculations by hand or with mechanical calculators. Large groups of human computers were used in astronomy, navigation, ballistics, engineering, and later for the numerical solution of differential equations. By the late 19th and early 20th centuries, many of these computing groups were composed largely of women. Examples include the astronomical computers working for the *Nautical Almanac* and later teams at observatories, government laboratories, and research institutions.

During World War II, the scale of numerical work *exploded* (pun intended). Hundreds of human computers computed ballistic firing tables and solved systems of finite-difference equations for artillery trajectories, fluid flow, and other military problems. Post-WWII, human computers were also instrumental to the US space program. If you've seen the movie *Hidden Figures*, you learned about some of these remarkable women who helped make going to the Moon possible by producing many of the calculations needed to determine flight plans, rocket burn parameters, launch windows, and safe re-entry trajectories.

Finite-difference methods were especially important because many physical problems could not be solved analytically. Human computers discretized differential equations onto grids and repeatedly computed numerical updates row by row—an extremely laborious process that helped motivate the development of electronic computers. The first six programmers of the world's first general-purpose electronic computer, ENIAC, were women who had previously worked as human computers.

David Alan Grier, **2005**, *When Computers Were Human*, Princeton University Press.
Kathy Kleiman, **2022**, *Proving Ground: The Untold Story of the Six Women Who Programmed the World's First Modern Computer.*
David Alan Grier, **2001**, *Human computers: the first pioneers of the information age*, **Endeavour** 25(1).

Setup

- Each student is assigned a position in a line.
- Each student is given an initial temperature value T_i.
- The two students at the ends of the line represent **fixed boundary conditions**.

Update rule

At each step, interior nodes update their temperature according to:

$$T_i^{\text{new}} = \frac{T_{i-1} + T_{i+1}}{2} \tag{3}$$

This means:

- you look at your two neighbors,
- take the average,
- and update your value.

Question 1. Prediction
Before starting, discuss:

- What happens to a sharp temperature spike over time?
- Will the system become smoother or less smooth?

Question 2. Iteration
Repeat the following for several steps:

1. Show your value to your neighbors.
2. Compute your new value.
3. Update simultaneously.

Record the evolution of the system. (Use the table below to track temperature values at each node across three time steps.)

Node \ Step	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
0										
1										
2										
3										
4										
5										
6										
7										
8										

Question 3. Observations
Discuss:

- Where do the largest changes occur?
- What happens to temperature peaks?
- How does heat move through the system?

Question 4. Interpretation

Explain in words:

- Why does the system become smoother over time?
- What role do neighboring nodes play?
- How does this relate to diffusion?

Optional Extension 1: Heat Production

Introduce a constant heat source at one node.

New update rule:

$$T_i^{\text{new}} = \frac{T_{i-1} + T_{i+1}}{2} + S \tag{4}$$

where S is a constant heat input.

Tasks:

- What happens to the system over time?
- Does it still become uniform?
- What steady-state behavior emerges?

Optional Extension 2: Two-Dimensional Grid

Arrange students in a grid.

Update rule:

$$T_{i,j}^{\text{new}} = \frac{T_{i+1,j} + T_{i-1,j} + T_{i,j+1} + T_{i,j-1}}{4} \tag{5}$$

Tasks:

- Observe how a hot region spreads in two dimensions.
- Compare to the one-dimensional case.
- Describe the shape of the temperature field over time.

Final Reflection

- In what sense are you "solving the heat equation"?
- Why does this method require many iterations?

Demo

There is a simple python GUI, `finitediff_gui.py`, that you can use to explore calculations similar to the ones we just did in this exercise. The implementation solves the Poisson equation:

$$\nabla^{2\mathrm{T}} = -\frac{A}{k}$$

using the discrete equation

$$T_i^{n+1} = \frac{k_{i-1} T_{i-1}^n + k_{i+1} T_{i+1}^n}{k_{i-1} + k_{i+1}} + \frac{A_i}{k_i}$$

Key Ideas

Heat flow is a local process: each point evolves based on its neighbors.

Numerical models evolve through local interactions. Each node only communicates with its immediate neighbors. Global behavior (e.g., a uniform steady state) emerges from many local interactions over many time steps. There is no global solution found in one step.

Information propagates at a finite rate. Because each update depends only on nearest neighbors, a temperature change at one end of the chain cannot instantaneously affect the other end. It must propagate node by node, step by step. This is the discrete analogue of finite diffusion speed in the continuum.

Steady state is reached when updates become zero. The system reaches a steady state when every node's temperature equals the average of its neighbors: $T_i = (T_{i-1} + T_{i+1})/2$. This is the discrete form of $\nabla^2 T = 0$ — the Laplace equation for heat without sources.

Activity 10.C
Boundary Conditions and Geological Meaning

Purpose

To understand how boundary conditions encode geological assumptions in numerical models.

A numerical model does not "know" geology—it only knows equations and boundary conditions.

Learning Outcomes

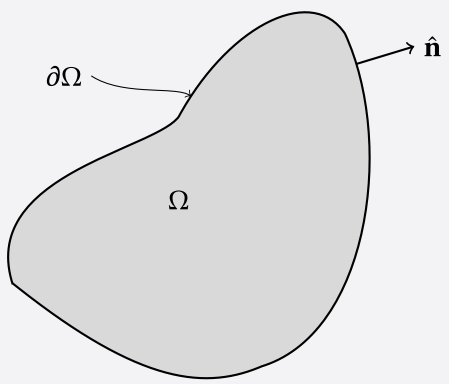
By the end of this activity, you should be able to:

- Distinguish between Dirichlet (fixed value) and Neumann (fixed flux) boundary conditions.
- Match appropriate boundary conditions to geological settings (ocean floor, continental lithosphere, cooling intrusion).
- Explain how changing a boundary condition changes the physical system the model represents.
- Argue why inappropriate boundary conditions can produce physically misleading results even when the numerical solution is mathematically stable.

Boundary conditions determine what kind of Earth the model represents.

Box .2: Boundary Conditions

A PDE describes how a field behaves *inside* a domain, Ω. To get a unique solution we must also specify how the domain interacts with its surroundings along the boundary, $\partial\Omega$. That extra information is the **boundary condition** and geologically, it is where an assumption about the outside world enters the model.



For a field $u(\mathbf{x})$, four standard conditions prescribe different combinations of the boundary value u and normal flux $\frac{\partial u}{\partial \hat{\mathbf{n}}}$ on $\partial\Omega$, with the first two types being most common and the second two being special cases:

Types of boundary conditions applied to domain boundaries.

Box .2: continued.

Type	Physical meaning	Prescribes on $\partial\Omega$	Example
Dirichlet	environment imposes fixed value at $\partial\Omega$	$u = f(\mathbf{x})$ (Type I)	Ocean floor: $T_{\text{surf}} = 4\,^\circ\text{C}$
Neumann	environment imposes fixed rate of energy, mass, current, etc. transfer across $\partial\Omega$	$\dfrac{\partial u}{\partial \hat{\mathbf{n}}} = g(\mathbf{x})$ (Type II)	Mantle heat flux entering the base of the lithosphere
Robin	process links value & flux at $\partial\Omega$	$au + b\dfrac{\partial u}{\partial \hat{\mathbf{n}}} = c$	Intrusion cooling by exchange with country rock: $-k\partial T/\partial n = h(T - T_{\text{melt}})$
Cauchy	$\partial\Omega$ strongly constrained by independent observations	$u = f(\hat{\mathbf{x}})$, $\dfrac{\partial u}{\partial \hat{\mathbf{n}}} = g(\mathbf{x})$	Borehole where both T and heat flow are measured

The governing equation describes the physics *inside* the domain; boundary conditions describe how the model interacts with the outside world. Two models with the same PDE and physical properties can produce completely different results if different boundary conditions are chosen — choosing a boundary condition is a geological modelling decision, not merely a mathematical one.

Question 1. Thermal evolution scenarios

Boundary conditions matter for evolving systems too, not just steady ones.

For each, state which boundary condition type you would use at the boundary relevant to its thermal evolution, and justify your choice physically.

Case	Upper BC	Lower BC	Why?
sedimentation/erosion			
cooling pluton			
hotspot			
cooling oceanic plate			
cratonic lithosphere			

Question 2. What the borehole tells you

The oceanic case fits the seafloor to a single condition: seawater keeps the seafloor close to a fixed temperature (Dirichlet). Continental lithosphere is different; a borehole gives you both a temperature and a gradient (hence heat flow) at the same location.

Which boundary condition type does that make? Referring back to the summary table, explain why the continental case cannot be reduced to a single fixed value or a single fixed flux the way the oceanic case can.

Heat flow is also measured at the seafloor, away from zones of active hydrothermal circulation, and is used to test age-dependent cooling models through time—*even though it is not imposed as a boundary condition there.* **Why is oceanic heat flow used to test the model rather than to set its boundary condition?**

Question 3. Extension — the lower boundary

This question is more open-ended than the others — there isn't a single clean answer, and that's the point.

For continental lithosphere, the base is often assigned a fixed temperature at the mantle adiabat. But unlike a borehole measurement, we don't know in advance how deep that boundary should sit.

If heat production $A(z)$ is specified through the crust, conservation of energy links the heat flow entering the base, q_b, to the heat flow measured at the surface, q_0:

$$q_0 = q_b + \int_0^{z_b} A(z)\,\mathrm{d}z$$

where z_b is the depth of the lower boundary.

Explain why fixing T to the mantle adiabat at the base makes this a harder problem than fixing T at a known depth. What quantity has to be solved for, rather than simply assumed?

Question 4. Critical thinking

Task: Explain why inappropriate boundary conditions can produce physically misleading results—even if the numerical solution is mathematically stable.

Reflection

If two models use the same equation but different boundary conditions, can they produce completely different results? Why?

Key Ideas

Boundary conditions are where geological assumptions enter a numerical model.

Boundary conditions define the physical system, not just the mathematics. Two models with identical equations but different boundary conditions represent completely different geological systems. Choosing a fixed temperature surface is equivalent to assuming an ocean floor in contact with circulating seawater; choosing a fixed heat flux is equivalent to assuming a continental surface where temperatures are free to evolve.

Dirichlet and Neumann conditions represent different physical constraints. A Dirichlet condition fixes the value (e.g., temperature); a Neumann condition fixes the gradient (e.g., heat flow across the surface). Robin and Cauchy boundary conditions are combinations of Dirichlet and Neumann conditions, either in linked or independent, respectively.

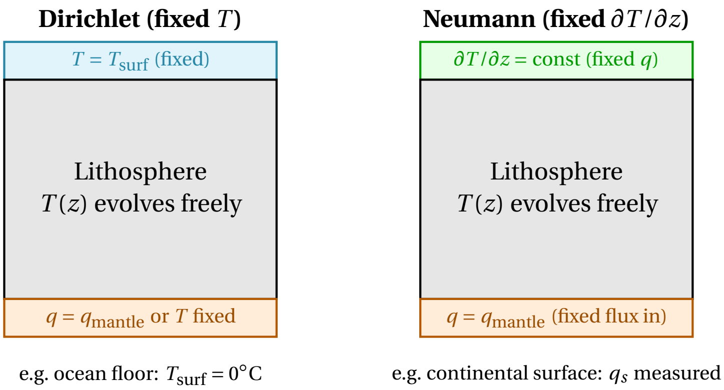


Figure 2: Dirichlet (left) fixes temperature at a boundary — appropriate where temperature is controlled externally (ocean floor). Neumann (right) fixes the heat flux gradient — appropriate where a flux is imposed but temperature is free to adjust (base of lithosphere, continental surface heat flow).

Geological understanding is required to choose appropriate boundary conditions. A numerically stable model with inappropriate boundary conditions can produce results that look reasonable but represent geology that does not exist. Interpreting model output always requires checking that the boundary conditions faithfully represent the geological setting being modelled.

Activity 10.D
Stability and Information Transfer

Purpose

To develop physical intuition for numerical stability in explicit finite difference schemes. In a time-stepping model, a field is updated step-by-step. Each step (iteration) uses information from neighboring nodes. Think of information as spreading gradually through the system.

Learning Outcomes

By the end of this activity, you should be able to:

- State the stability condition for the explicit finite difference heat equation and identify the quantities it depends on.
- Explain physically why the allowable time step is proportional to Δz^2 and inversely proportional to κ.
- Describe what physically happens when the stability limit is exceeded, and why this corresponds to physically impossible behavior.
- Compare stability requirements for materials with different thermal diffusivities.

Box .3: Diffusion as a Finite Difference Problem

The 1-D diffusion equation for temperature $T(z,t)$ is

$$\frac{\partial T}{\partial t} = \kappa \frac{\partial^2 T}{\partial z^2},$$

with κ the thermal diffusivity. Discretize space into nodes $T_i \approx T(i\,\Delta z, t)$ and time into levels $T_i^n \approx T(i\,\Delta z, n\,\Delta t)$. The spatial second derivative is approximated with the usual central difference,

$$\left.\frac{\partial^2 T}{\partial z^2}\right|_i \approx \frac{T_{i+1} - 2T_i + T_{i-1}}{\Delta z^2},$$

and we define the dimensionless stability parameter

$$r = \frac{\kappa\,\Delta t}{\Delta z^2}.$$

The three schemes differ only in *which time level* the right-hand side is evaluated at.

Explicit (forward-time, centred-space, FTCS)

Evaluate the spatial derivative at the *old* time level n:

$$\frac{T_i^{n+1} - T_i^n}{\Delta t} = \kappa \frac{T_{i+1}^n - 2T_i^n + T_{i-1}^n}{\Delta z^2}.$$

Solving directly for T_i^{n+1}:

$$T_i^{n+1} = T_i^n + r\left(T_{i+1}^n - 2T_i^n + T_{i-1}^n\right).$$

Every term on the right is already known, so this is an explicit update — no system of equations to solve, but (as derived earlier) stable only for $r \le \frac{1}{2}$.

Implicit (backward-time, centred-space, BTCS)

Box .3: continued.

Evaluate the spatial derivative at the *new* time level $n+1$ instead:

$$\frac{T_i^{n+1} - T_i^n}{\Delta t} = \kappa \, \frac{T_{i+1}^{n+1} - 2T_i^{n+1} + T_{i-1}^{n+1}}{\Delta z^2}.$$

Rearranging so all unknowns $(n+1)$ are on the left and all knowns (n) are on the right:

$$-r\,T_{i-1}^{n+1} + (1+2r)\,T_i^{n+1} - r\,T_{i+1}^{n+1} = T_i^n.$$

This couples every node's new value to its neighbors' new values, so it is a tridiagonal linear system solved simultaneously for the whole row — the cost of unconditional stability is a linear solve every time step, not just a formula evaluation.

Crank–Nicolson: averaging the two

Crank–Nicolson evaluates the spatial derivative as the *average* of the explicit (old-level) and implicit (new-level) versions:

$$\frac{T_i^{n+1} - T_i^n}{\Delta t} = \frac{\kappa}{2}\left[\frac{T_{i+1}^n - 2T_i^n + T_{i-1}^n}{\Delta z^2} + \frac{T_{i+1}^{n+1} - 2T_i^{n+1} + T_{i-1}^{n+1}}{\Delta z^2}\right].$$

Rearranged into the same known/unknown form as implicit:

$$-\frac{r}{2}\,T_{i-1}^{n+1} + (1+r)\,T_i^{n+1} - \frac{r}{2}\,T_{i+1}^{n+1} = \frac{r}{2}\,T_{i-1}^n + (1-r)\,T_i^n + \frac{r}{2}\,T_{i+1}^n.$$

Like implicit, this is a tridiagonal system each step.

Setup

In this activity, we will will be working with a simple thermal model of a temperature spike in a background thermal gradient. To illustrate some of the important concepts without the complications of a specific unit system, we will use the units summarized in the table below.

Quantity	Label
Δz	length units (L)
Δt	time units (T)
κ	L^2/T
T	degrees $^\circ$
$r = \kappa \frac{\Delta t}{\Delta z^2}$	dimensionless

We will be using Dirichlet boundary conditions with fixed temperature of 15 $^\circ$ at the surface and 55 $^\circ$ at the base. For simplicity, we are assuming a thermal diffusivity $\kappa = 1\,L^2\,T^{-1}$. Our grid is discretized with an interval of $\Delta z = 2\,L$.

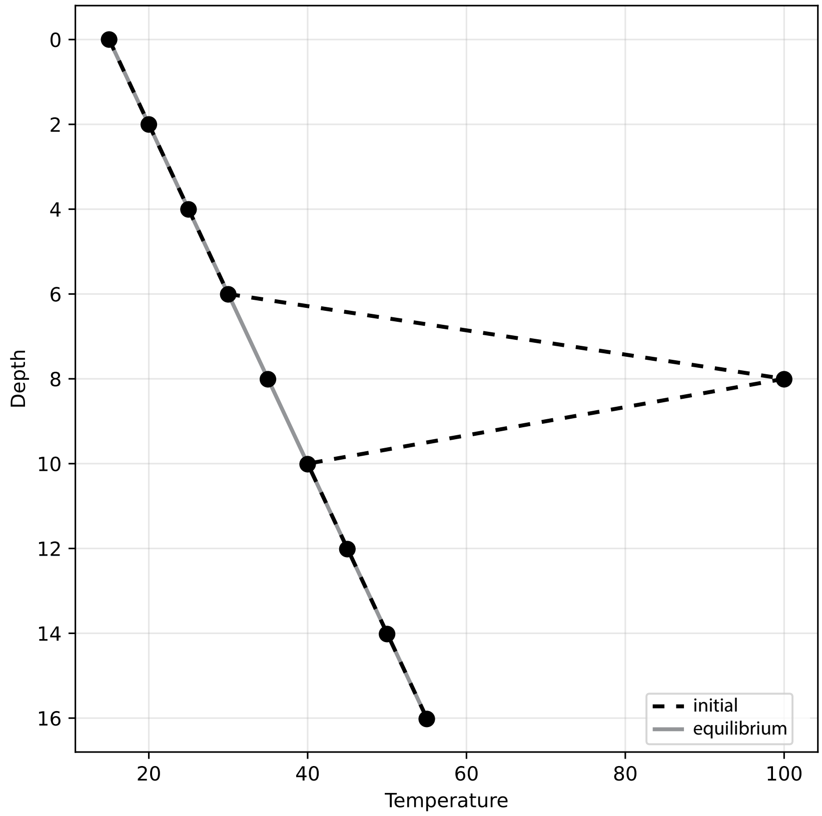


Figure 3: Initial and equilibrium temperatures for a used to diffusion models for stability.

Tasks

Complete the following questions:

Question 1. Prediction and reasoning
Before running anything, discuss:

- What do you expect for a very small time step? A very large one?
- Heat diffuses at a rate set by κ. Why can't it move arbitrarily far in a single time step?

Question 2. What breaks, and when

Table 1: First three time steps using the explicit method, with differing Δt.

			$\Delta t = 1$ T			$\Delta t = 1.75$ T			$\Delta t = 2.25$ T			$\Delta t = 3$ T		
Depth L	Initial T ∘	t[T] =	5.2 ∘	10.4 ∘	16 ∘	5.25 ∘	10.5 ∘	17.5 ∘	4.5 ∘	11.2 ∘	18 ∘	6 ∘	12 ∘	18 ∘
0	15		15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0
2	20		22.8	23.8	23.3	25.4	23.3	23.0	20.0	36.4	2.2	20.0	-34.8	-234
4	25		32.6	32.5	31.1	29.7	33.7	31.0	45.6	8.3	69.3	61.6	162	530
6	30		43.3	40.3	38.1	47.7	38.5	37.2	20.9	72.1	-13.4	-18.8	-159	-648
8	100		51.0	46.4	43.8	44.5	47.6	43.5	77.1	9.8	98.0	124	272	799
10	40		53.3	50.3	48.1	57.7	48.5	47.2	30.9	82.1	-3.4	-8.8	-149	-638
12	45		52.2	52.5	51.1	49.7	53.7	51.0	65.6	28.3	89.3	81.6	182	550
14	50		52.6	53.8	53.3	55.4	53.3	53.0	50.0	66.4	32.2	50.0	-4.8	-204
16	55		55.0	55.0	55.0	55.0	55.0	55.0	55.0	55.0	55.0	55.0	55.0	55.0

In Table **??** (above) are the explicit finite-difference solution to a temperature spike in a uniform thermal gradient (Figure **??**). In the table below, record what you observe for each run (smooth decay, mild wobble, or blow-up), then compute $\Delta t/\Delta z^2$ for each case.

Δt	Δz	$\Delta t/\Delta z^2$	**Solution behavior**
1	2		
1.75	2		
2.25	2		
3	2		

Looking at your table, what value of $\Delta t/\Delta z^2$ seems to separate stable runs from unstable ones?

Question 3. Reading the stability condition

For a explicit scheme, the stability criteria requires that the grid dimensions and step size satisfy the inequality

$$\kappa \frac{\Delta t}{\Delta z^2} \leq \frac{1}{2d},$$

where d is the dimensionality, 1–3. If we rearrange this equation into the parts we can change and those that can't,

$$2d\kappa \leq \frac{\Delta z^2}{\Delta t}.$$

Compare this to the threshold you found in Question 2. **Explain** why the allowable time step depends on Δz and κ the way it does—what does this mean physically, in terms of how far thermal information can travel between nodes in one step?

Table 2: First three time steps using the implicit method, with differing Δt.

			$\Delta t = 1$ T			$\Delta t = 1.75$ T			$\Delta t = 2.25$ T			$\Delta t = 3$ T		
Depth L	Initial T °	t[T] =	5.2 °	10.4 °	16 °	5.25 °	10.5 °	17.5 °	4.5 °	11.2 °	18 °	6 °	12 °	18 °
0	15		15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0	15.0
2	20		22.1	23.3	23.2	22.1	23.2	23.1	21.7	23.2	23.0	22.1	23.1	23.0
4	25		31.1	32.1	31.2	30.9	31.9	30.9	30.1	31.7	30.8	30.7	31.5	30.8
6	30		43.1	40.7	38.3	42.7	40.6	38.0	42.5	40.3	37.9	42.1	40.1	38.0
8	100		54.4	47.4	44.2	55.2	47.4	43.8	58.6	47.1	43.7	55.3	47.0	43.8
10	40		53.1	50.7	48.3	52.7	50.6	48.0	52.5	50.3	47.9	52.1	50.1	48.0
12	45		51.1	52.1	51.2	50.9	51.9	50.9	50.1	51.7	50.8	50.7	51.5	50.8
14	50		52.1	53.3	53.2	52.1	53.2	53.1	51.7	53.2	53.0	52.1	53.1	53.0
16	55		55.0	55.0	55.0	55.0	55.0	55.0	55.0	55.0	55.0	55.0	55.0	55.0

In Table **??** (above) the temperature spike experiment has been repeated using the implicit method instead. Record what you observe for each run (smooth decay, mild wobble, or blow-up), then compute $\Delta t/\Delta z^2$ for each case.

Δt	Δz	$\Delta t/\Delta z^2$	**Solution behavior**
1	2		
1.75	2		
2.25	2		
3	2		

You may notice that this time all the runs are stable. Implicit solutions are unconditionally stable. There is a third formulation, much like the forward (explicit), and backwards (implicit) there is method similar to the central difference in time called the Crank–Nicholson Method. It involves taking the average of the explicit and implicit solutions wrapped up into one and is solved with a tridiagonal solver in the same way as the implicit method. This method is more accurate and also stable. Why is this method more accurate?

Question 4. Physical reasoning

Interpret the stability condition as a limit on how fast thermal information can move between nodes.

Why can heat not “jump” over nodes in a single time step?

What determines how quickly heat diffuses through a material?

Key Ideas

Numerical instability occurs when the model allows heat to propagate faster than physically possible. Heat diffuses at a rate set by κ. In a time step Δt, physical heat can travel at most $\sqrt{2\kappa\,\Delta t}$. The stability condition $\Delta t < \Delta z^2/(2d\kappa)$ ensures that this thermal length cannot exceed the node spacing Δz. Violating the condition means asking the model to move heat further in one step than the physics allows.

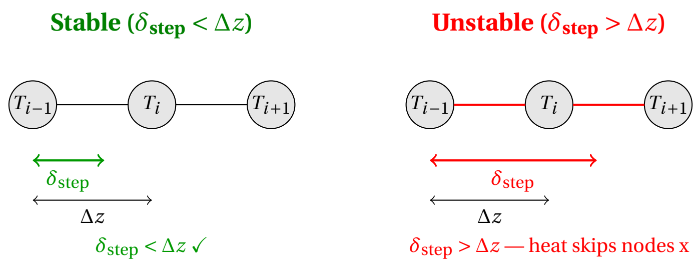


Figure 4: Stability requires that the thermal $\delta_{\text{step}} = \sqrt{2\kappa\,\Delta t}$ does not exceed the node spacing Δz. When it does, heat is asked to propagate past neighboring nodes in one step, which is physically impossible and causes numerical blow-up.

Finer grids require smaller time steps. Because stability scales as Δz^2, halving the grid spacing reduces the maximum allowable time step by a factor of four. Increasing resolution is therefore expensive: the number of spatial nodes doubles *and* the number of time steps quadruples to cover the same total time.

High diffusivity is more demanding numerically. Materials with large κ (e.g., peridotite, ice) diffuse heat quickly and require smaller time steps for numerical stability than low-diffusivity materials (e.g., clay-rich sediments). This is the reverse of the geological situation: fast-diffusing materials are easier to model geologically but harder numerically.